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FULL-STACK E-COMMERCE BLUEPRINT.

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Project Architecture.

A full-stack fashion e-commerce web application engineered as a university assignment.



Brand Identity.

Simulating a real-world digital storefront with bold, high-energy aesthetics inspired by Zara and Nike.

Graphic Tee

\$45

NotebookLM

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The Design System.

Utilizing strong sans-serif typography alongside a pink-dominant brand identity.

UI Component Matrix.

Engineered with full-width banners, sticky navigation, pill-shaped conversion buttons, and floating cart icons.

Cropped Hoodie

\$65

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Dark Mode Execution.
Base flain black background
paired with magenta and
white typography.



Dark Mode Execution.
Base deep black background
paired with magenta and
white typography for
maximum contrast.



Light Mode Transformation.
Base off-white paired with
dusty rose and near-black
text, maintaining core brand
identity while shifting
atmospheric contrast.

High-Waist Jeans

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The Landing Engine.
Homepage structured around dynamic full-width hero banners, featured collections, and trending products.

Dynamic Catalog Grid.
Responsive HTML/CSS flex-grids featuring real-time data filtering by category, size, color, and price thresholds.

Oversized Jacket

\$89

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Product Detail Architecture
Rendering multi-angle image views, stateful size selection, and reactive quantity pickers.



Conversion Drivers
Built around bold 'Add to Cart' interactions and algorithmic related-product rendering.



Athleisure Set

\$55

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Cart State Management

Live-updating price summaries, dynamic item removal, and real-time quantity controls.

The Checkout Pipeline

Multi-step form UI capturing shipping, mock payment gateways, and final order confirmation rendering.

Streetwear Dress

\$95

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Authentication Protocol

Secure Login/Signup routing powered by JWT-based authentication tokens.



Role-Based Security

Clean form UI branching access states between standard consumer shopping sessions and elevated system administrators.

Midnight Cargo Pants

\$120
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The Protected Dashboard.

Admin panel featuring full product CRUD management, inventory counts, and global sales metrics.



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Admin panel featuring full product CRUD management, inventory counts, and global sales metrics.



User Order Tracking

Real-time pipeline visibility allowing users to trace state updates:
Placed → Processing → Shipped → Delivered.

Urban Tactics Jacket

\$120

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The Backend Engine.
Built on Node.js and Express.js, engineering a robust RESTful API pipeline.



Data Bridging.
Seamlessly connecting front-end state changes to secure database mutations for users, products, and dynamic carts.

Street Runner Hoodie

\$85

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NoSQL Data Architecture

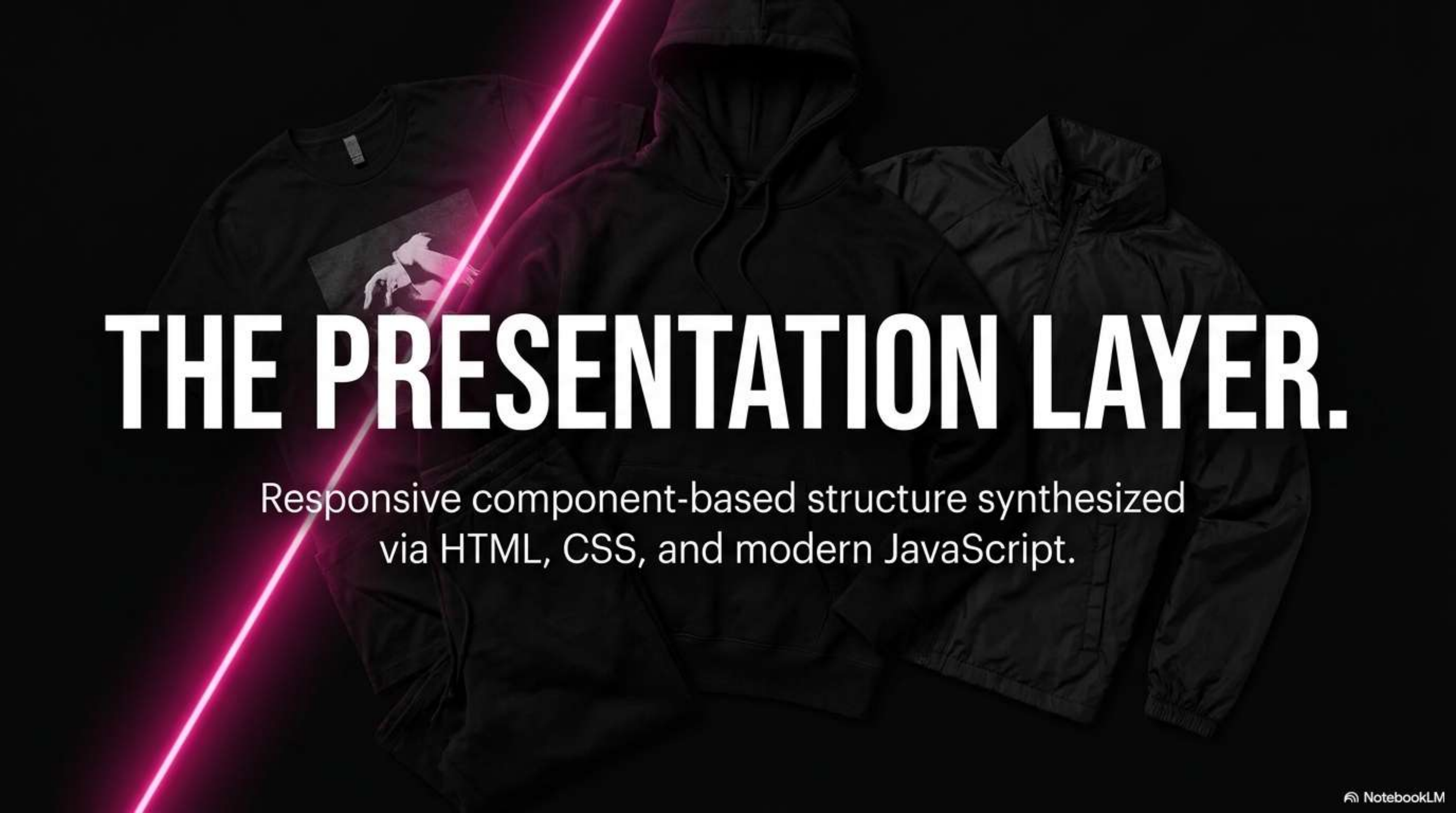
Utilizing MongoDB for scalable, non-relational document storage.

Collection Schemas.

Dedicated, interconnected database collections architected specifically for Users, Products, Cart states, and Order histories.

Reflective Jacket

\$120



THE PRESENTATION LAYER.

Responsive component-based structure synthesized
via HTML, CSS, and modern JavaScript.

A black tactical jacket with a high collar, multiple pockets, and a drawstring hood. It is centered in the frame and surrounded by a vibrant, glowing pink aura that fades into the dark background.

THE LOGIC LAYER.

RESTful routing pipelines powered by Node.js and Express.js,
brokering secure access to MongoDB collections.



THE CLOUD EDGE

Production-ready deployment via Vercel,
ensuring high-availability global delivery
and seamless continuous integration.



THE FULL-STACK SYNTHESIS.

Demonstrating hands-on mastery of frontend UX, backend API design, database integration, and role-based access control within a live e-commerce ecosystem.